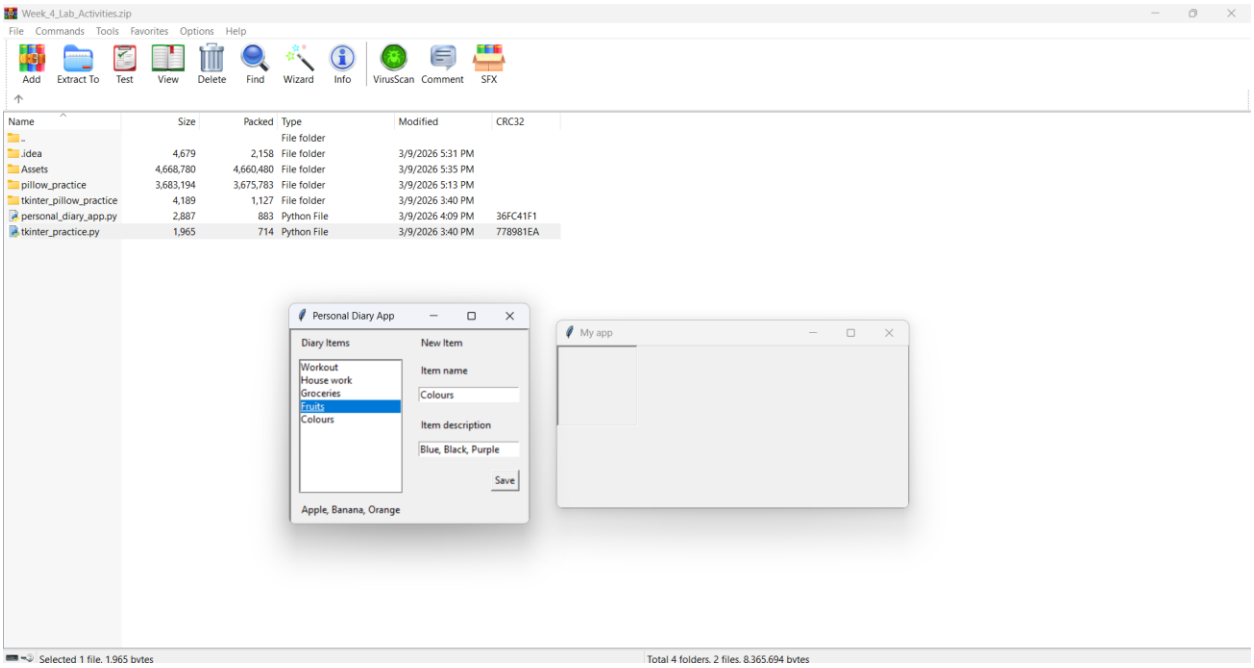
Software Technology 1 [4483] - Week_4_lab_report

u3306801

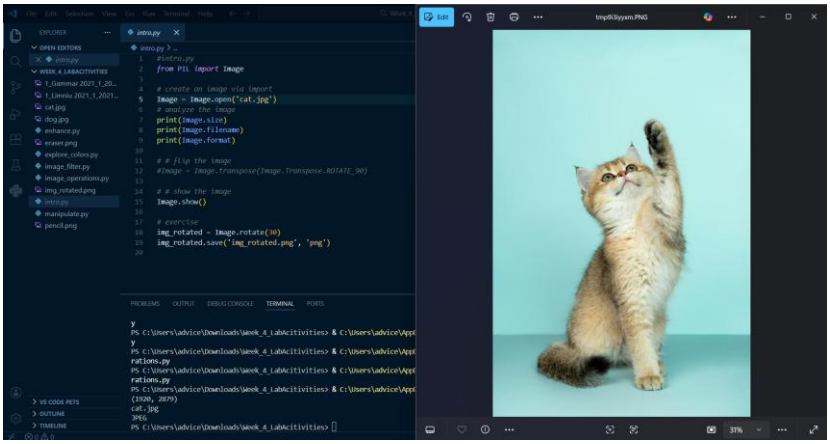
Task 1: Introduce to Tkinter

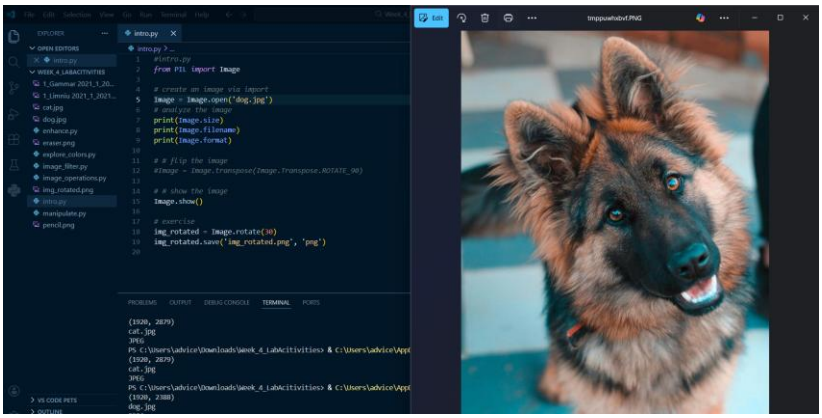
In this task, I ran the Python files tkinter_practice.py and personal_diary.py from week_4_Lab_Activites folder. The diary app shows a list of the items on the left such as workout, house work and groceries. On the right, there are boxes to type a new items name and description, and allow to save. This task shows hat the Tkinter library can be create simple GUI applications in Python.

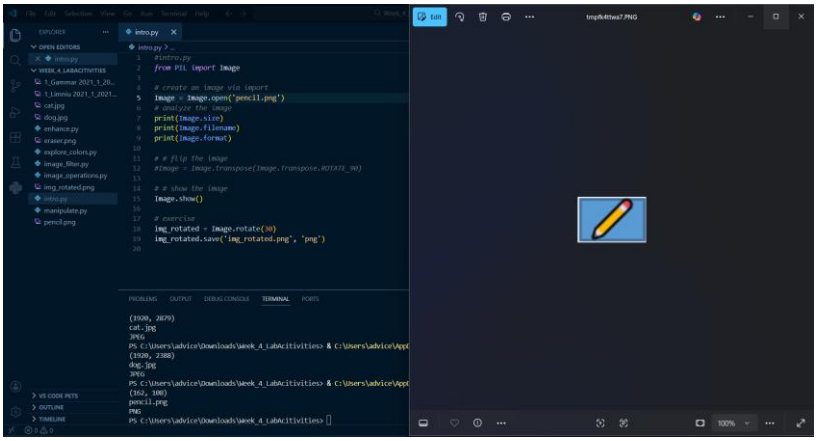


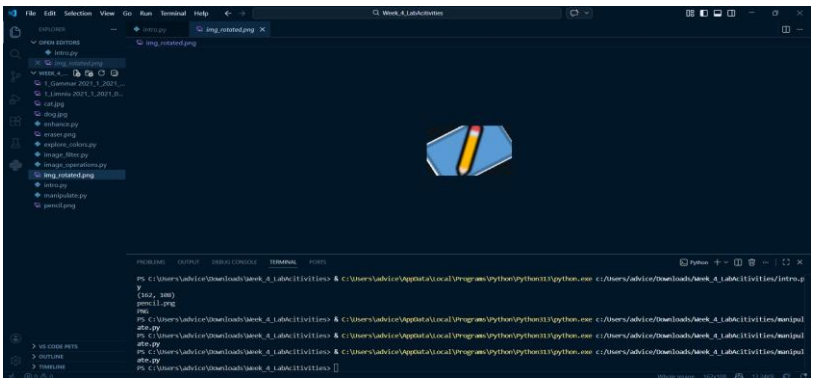
Task 2: Intro.py

In this task, I learned how to open an image by using the pillow library. I tested the script with three insect pictures. The code opened each image, rotated image by 30 degrees and save a new file. When I ran it, the rotated image looked slightly titled, and a new JPN and PNG file appeared in my folder. In addition, when I tested a different image, the program asked me to choose the folder again and save the file. This did not happen in the other tasks.


intro.py
#intro.py
from PIL import Image
# create an image via import
Image = Image.open('cat.jpg')
# analyze the image
print(Image.size)
print(Image.filename)
print(Image.format)
# # flip the image
#Image = Image.transpose(Image.Transpose.ROTATE_90)
# # show the image
Image.show()
# exercise
img_rotated = Image.rotate(30)
img_rotated.save('img_rotated.png', 'png')
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(1920, 2879)
cat.jpg
JPEG
PS C:\Users\advice\Downloads\week_4_LabActivities>

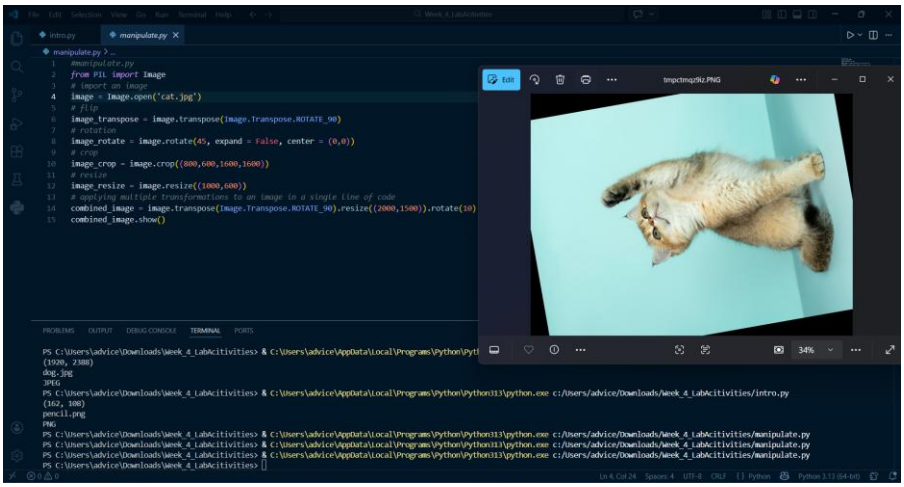

intro.py
#intro.py
from PIL import Image
# create an image via import
Image = Image.open('dog.jpg')
# analyze the image
print(Image.size)
print(Image.filename)
print(Image.format)
# # flip the image
#Image = Image.transpose(Image.Transpose.ROTATE_90)
# # show the image
Image.show()
# exercise
img_rotated = Image.rotate(30)
img_rotated.save('img_rotated.png', 'png')
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(1920, 2879)
cat.jpg
JPEG
(1920, 2388)
dog.jpg

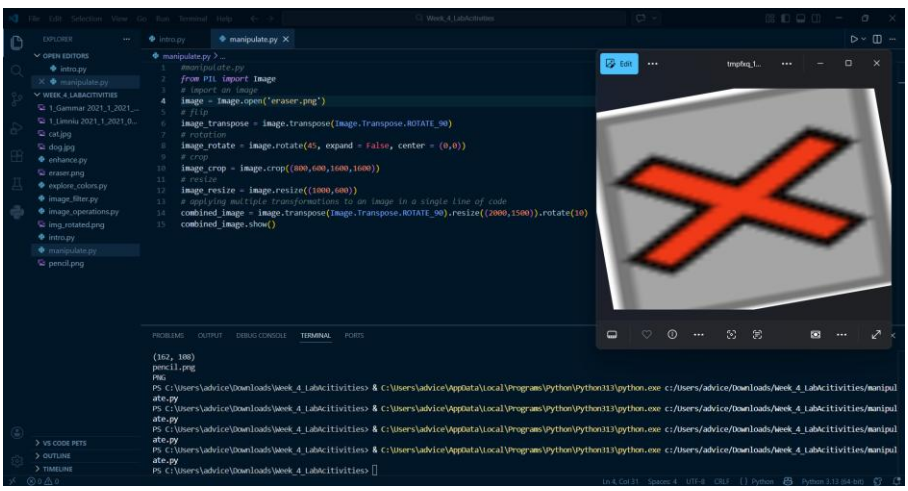

intro.py
#intro.py
from PIL import Image
# create an image via import
Image = Image.open('pencil.png')
# analyze the image
print(Image.size)
print(Image.filename)
print(Image.format)
# # flip the image
#Image = Image.transpose(Image.Transpose.ROTATE_90)
# # show the image
Image.show()
# exercise
img_rotated = Image.rotate(30)
img_rotated.save('img_rotated.png', 'png')
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(1920, 2879)
cat.jpg
JPEG
(1920, 2388)
dog.jpg
JPEG
(362, 100)
pencil.png
PNG

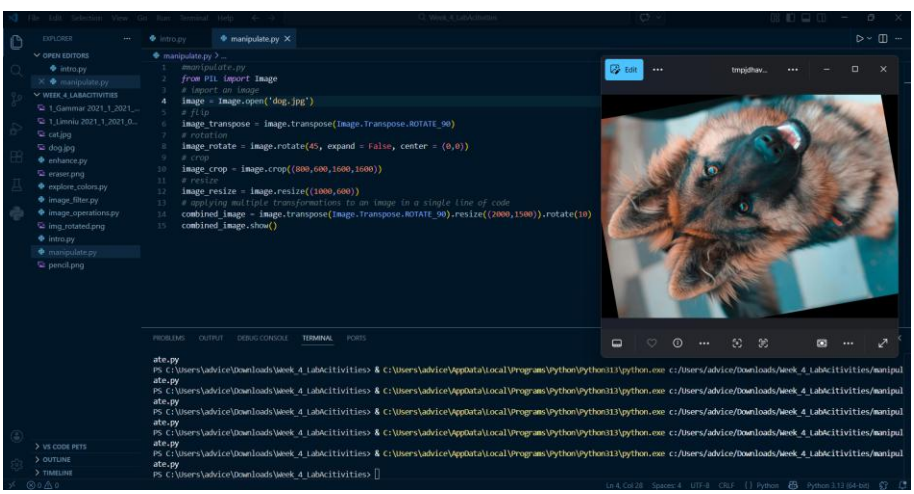

Week_4_LabActivities
intro.py
img_rotated.png
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(362, 100)
pencil.png
PNG

Task 3: Manipulate.py

I used a given script to change the image ways like flipping the picture, cropping it and resizing. I tested each change one by one, and the new image showed up correctly each time and the version with all the changes looked very edited.

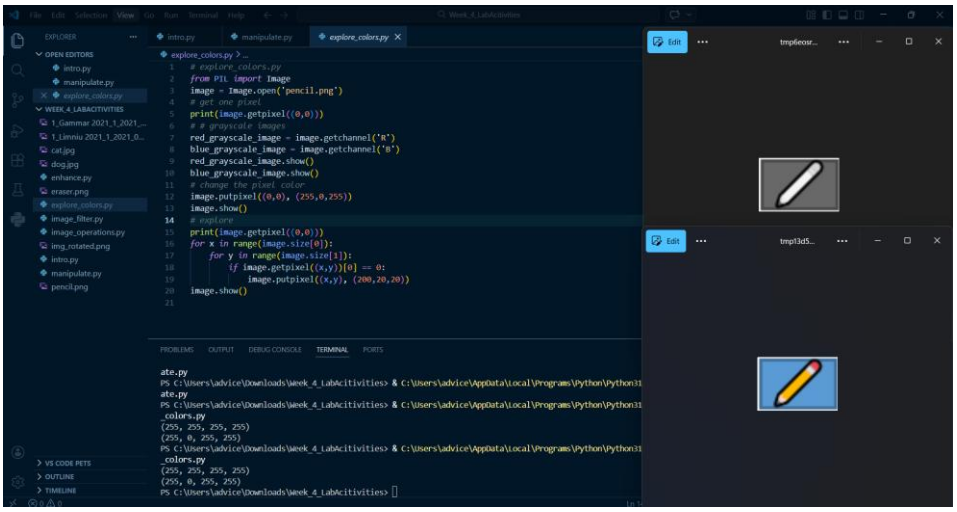


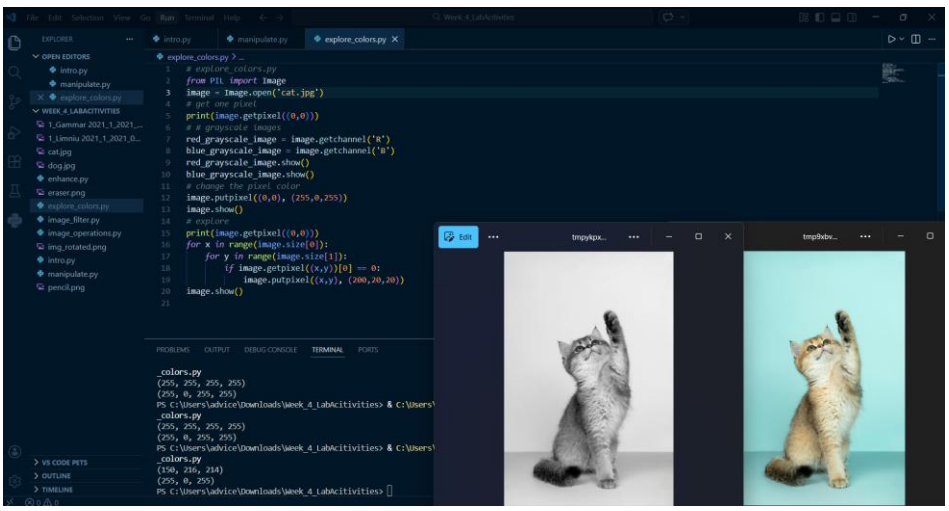


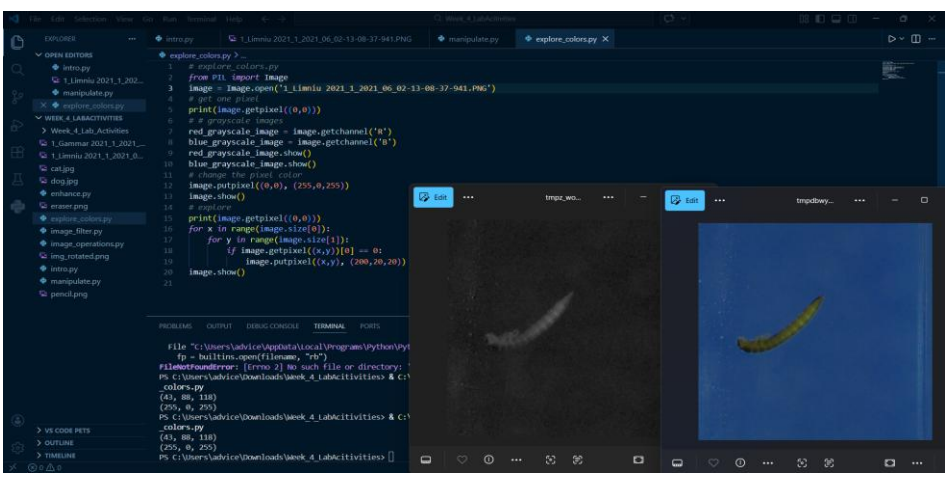


Task 4: explore_colours.py

I learned that images are made of tiny pixels and each pixel has an RGB color value (red, green, blue). the program used getpixel() to see the color of a pixel and putpixel() to change the color of some pixels. After changing the colors, the new image was shown successfully.

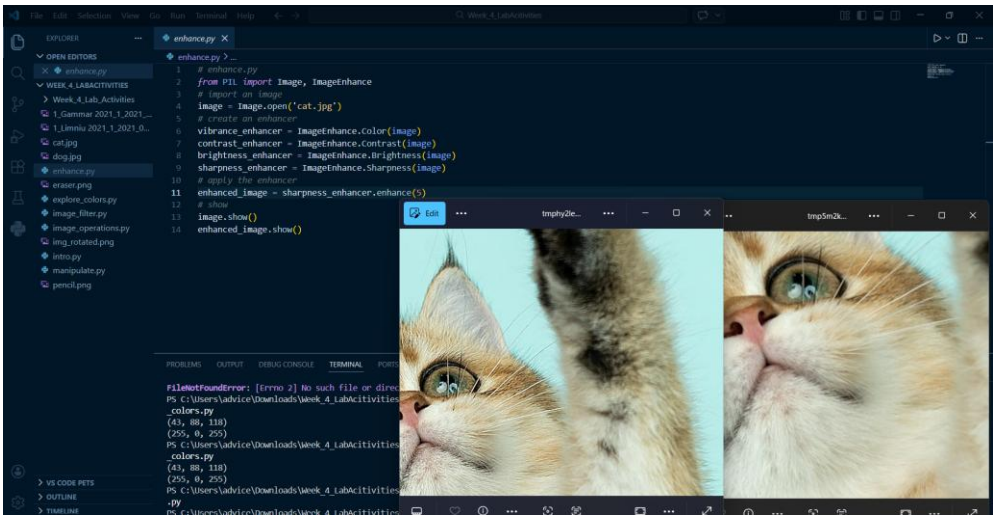


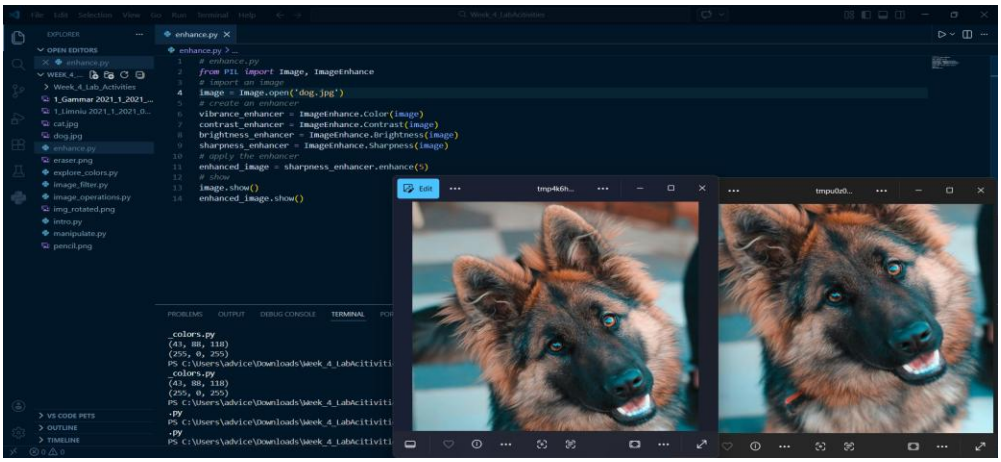


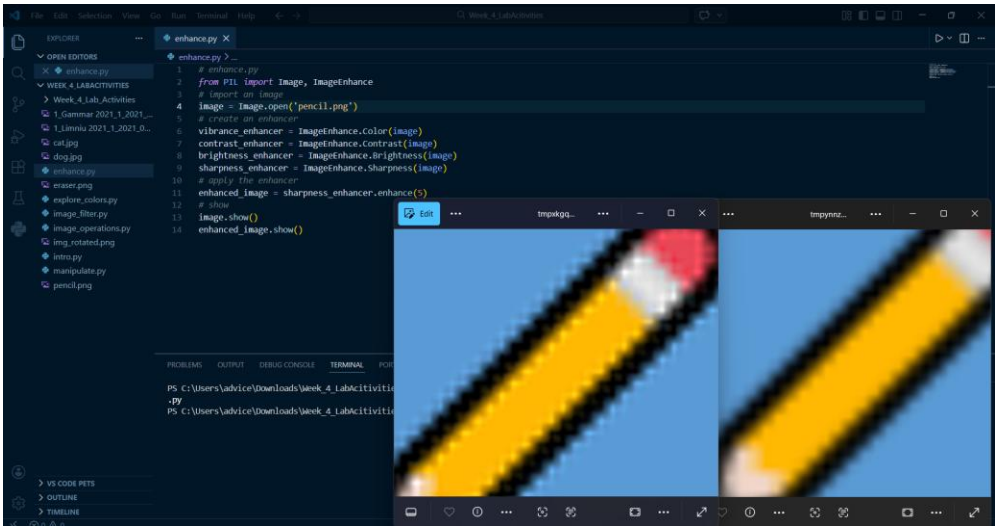


Task 5: enhance.py

In this task shows how to make images look clearer or strong using the ImageEnhance module. I tested color, contrast, brightness and sharpness on images. it looked brighter, more vibrant and more detailed compared to the original image.

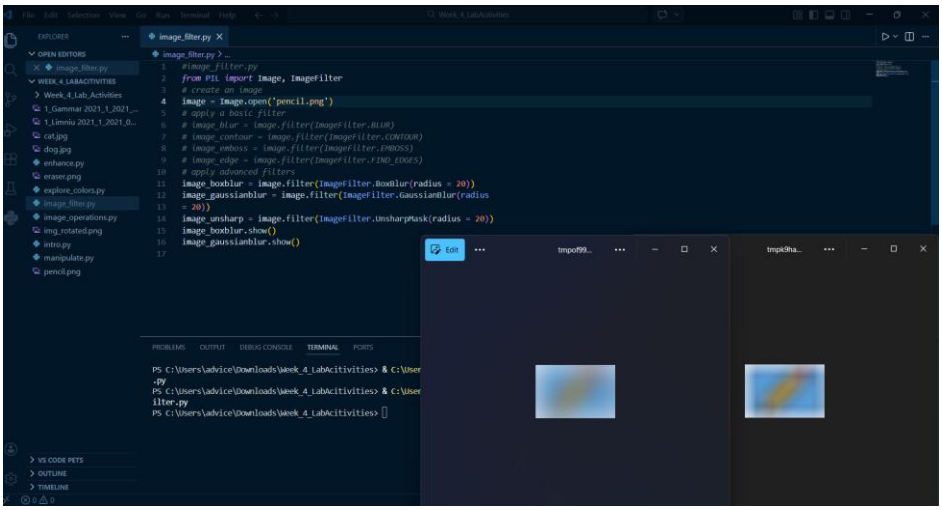


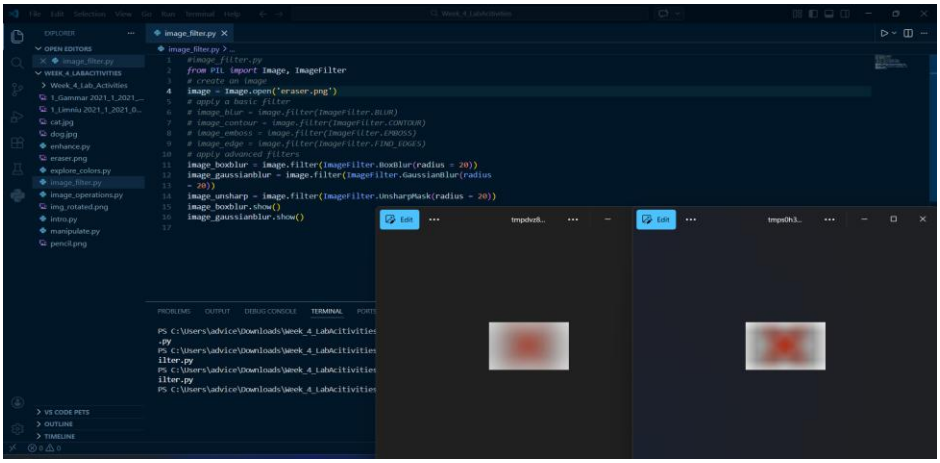


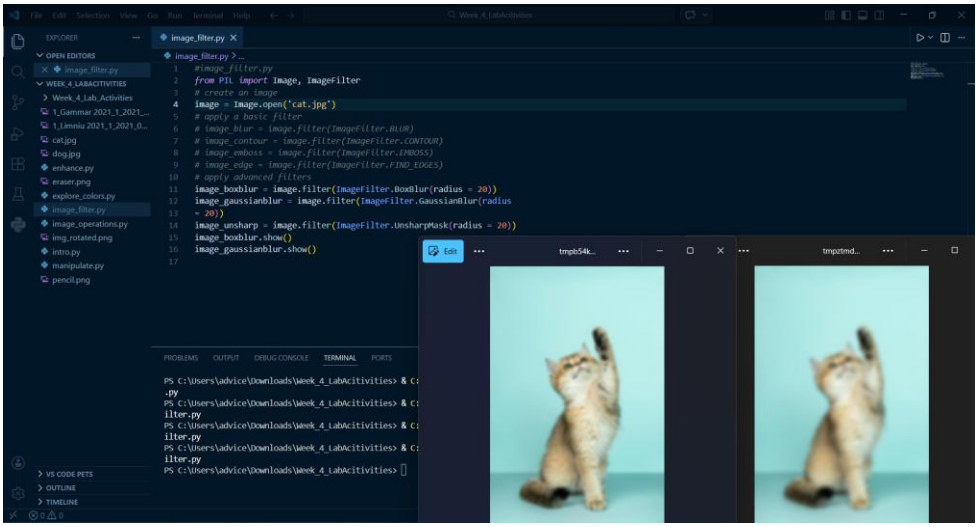


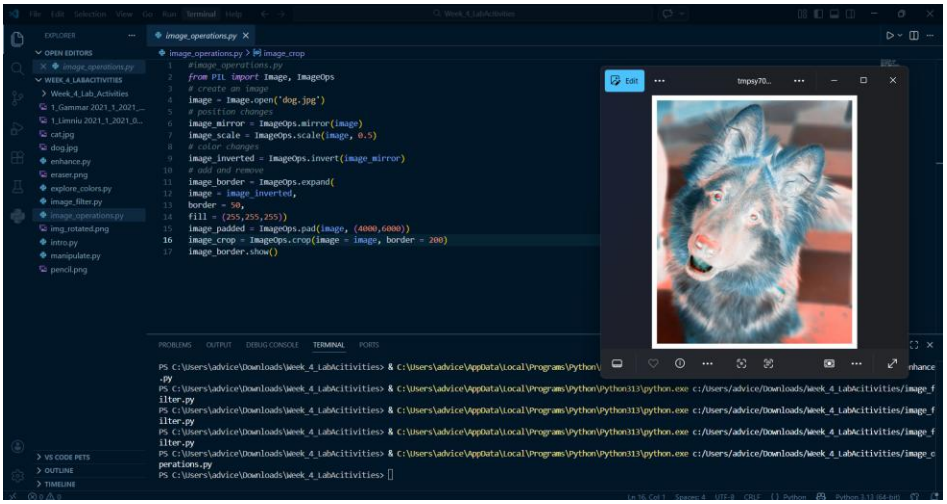
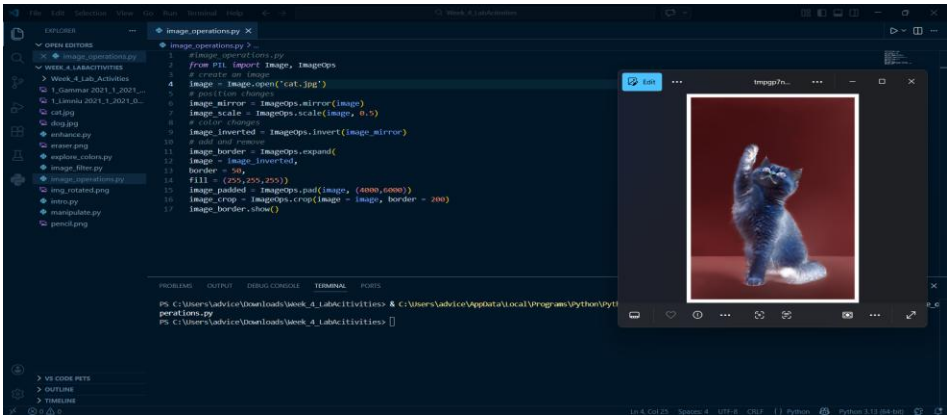
Task 6: Image_filtering.py

The program used filters like box blur and gaussian blur make the image look softer or smoother. These filters change the picture looks, also it often used in image editing and processing. The script worked well and the filter images were shown correctly.







Task 7: Image_operations.py

In this task was tasted by using given script to try tools from the ImageOps module. I tasted mirroring the image, investing the colors, and adding a border around it. The image change properly and looked different from the original image. Additionally, it cannot be run the given script if it's png file.